



**B.M.S. COLLEGE OF ENGINEERING,
BENGALURU-19**
Autonomous Institute, Affiliated to VTU
DEPARTMENT OF MATHEMATICS

THIRD SEMESTER B.E COURSE(CSE/ISE/CSE-AIDS/CS-IOT/CSE-DS)

Course Title: Statistics and Discrete Mathematics

Basic concepts

1. Define the following with an example.
(i) digraph (ii) graph (iii) order and size of graphs and digraphs (iv) loops and directed loops
(v) parallel/ multiple edges (vi) simple graph (vii) complete graph (viii) complete bipartite graph (ix) regular graph (x) 3-D hypercube
2. Suppose that in a group of 5 people: A, B, C, D, and E, the following pairs of people are acquainted with each other- A and C, A and D, B and C, C and D & C and E.
(a) Draw a graph G to represent this situation. (b) Identify the degree of each vertex.
3. Model the following situation as a graph and write its vertex set and the edge set. Also, write the degree of each vertex. In Netherland, there is a highway from *Amsterdam* to *Breda*, another highway from *Amsterdam* to *Capelle aan den IJssel*, a highway from *Breda* to *Dordrecht*, a highway from *Breda* to *Ede* and another one from *Dordrecht* to *Ede*, and a highway from *Capelle aan den IJssel* to *Ede*.
4. Let P, Q, R, S and T represent 5 cricket teams. Suppose that the teams P, Q, R have played one game with each other and the teams P, S, T have played one game with each other. Represent this situation as a graph and hence determine:
(i) The teams that have not played with each other.
(ii) The number of games played by each team.

Handshaking property

5. For a graph $G = (V, E)$, determine the largest possible value of $|V|$ if
(a) $|E| = 35$ and $\deg(v) \geq 3, \forall v \in V$
(b) $|E| = 19$ and $\deg(v) \geq 4, \forall v \in V$
6. For a graph with n vertices and m edges if δ is the minimum and Δ is the maximum of the degrees of the vertices, show that $\delta \leq \frac{2m}{n} \leq \Delta$.
7. Determine the order of the graph in the following cases:

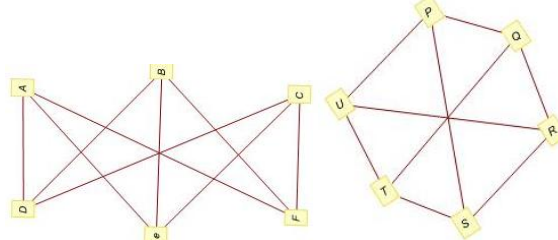
- (i) G is a cubic graph with 9 edges.
- (ii) G has 10 edges with 2 vertices of degree 4 and all other vertices of degree 3.
- (iii) G is a regular graph with 15 edges.
- 8. If $G = (V, E)$ is a simple graph, prove that $2|E| \leq |V|^2 - |V|$.
- 9. Show that a simple graph of order $n = 4$ and size $m = 7$ and a complete graph of order $n = 4$ and size $m = 5$ do not exist.
- 10. Let G be a graph of order 9 such that each vertex has degree 5 or 6. Prove that at least 5 vertices have degree 6 or at least 6 vertices have degree 5.
- 11. Show that there is no graph with 28 edges and 12 vertices such that:
 - (i) the degree of a vertex is either 3 or 4
 - (ii) the degree of a vertex is either 3 or 6
- 12. Show that every simple graph order greater than or equal to 2 must have at least 2 vertices of the same degree.

Isomorphism of graphs

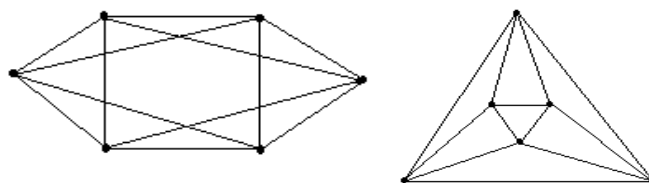
13. Define Isomorphism of graphs. Verify whether the following two graphs are isomorphic or not



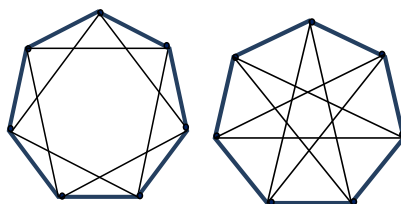
14. Define Isomorphism of the graph. Verify that the given two graphs are isomorphic or not.



15. Verify whether the following graphs are isomorphic or not.



16. With proper labeling show that the following graphs are isomorphic.



Subgraphs, Connected, Disconnected, Eulerian and Hamiltonian graphs, Trees

17. Define the following with an example: (i) subgraph (ii) spanning subgraph (iii) induced subgraph (iv) vertex and edge disjoint subgraphs (v) walk (vi) open and closed walks (vii) trail and circuit (viii) path and cycle (ix) connected and disconnected graphs (x) Eulerian graph (xi) Hamiltonian graph (xii) tree (xiii) spanning trees (xiv) minimal spanning tree

18. Suppose a committee has seven members, these members meet each day for lunch at a round table. They decide to sit in such a way that every member has different neighbors at each lunch.

How many ways can this arrangement last?

19. Let G be a simple graph with n vertices and m edges where m is at least 3. If

$$m \geq \frac{(n-1)(n-2)}{2} + 2, \text{ prove that } G \text{ is Hamiltonian graph.}$$

20. Prove that a connected graph G remains connected even after removing an edge e from G if and only if e is a part of some cycle in G .

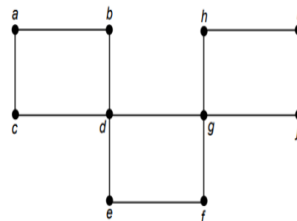
21. Let G be a disconnected graph of even order n with two components each of which is complete.

Prove that G has a minimum of $\frac{n(n-2)}{4}$ edges.

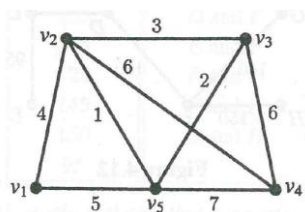
22. If G is a simple graph with n vertices in which the degree of every vertex is at least $\frac{(n-1)}{2}$, prove that G is connected.

23. Show that a connected graph with exactly 2 vertices of odd degree has an Euler trail.

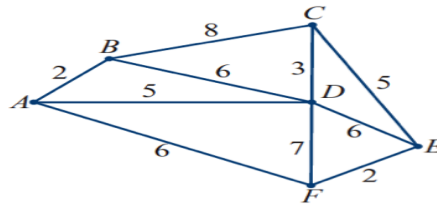
24. Find all the possible spanning trees of the given graph:



25. For the given graph, draw: (i) a spanning tree (ii) a pair of edge disjoint subgraphs (iii) the induced subgraph $\langle v_1, v_3, v_4, v_5 \rangle$



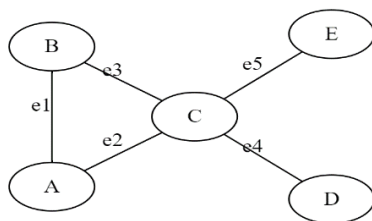
26. For the given graph, draw: (i) a spanning tree (ii) a pair of edge disjoint subgraphs (iii) the induced subgraph $\langle B, C, E, F \rangle$



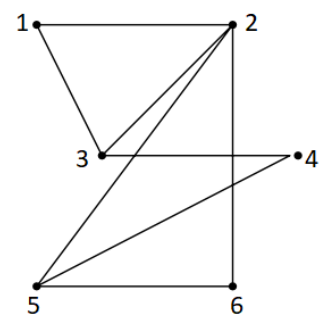
Matrix representations of a graph

27. For the given graph, derive its incidence and adjacency matrices. Hence, write any three observations each from these matrices:

(a)



(b)



28. Draw the graph G whose incidence matrix is given and hence obtain the adjacency matrix of the corresponding graph G . Also write any three observations on the adjacency matrix.

(a)

$$A(G) = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

(b)

$$A(G) = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

29. For the given adjacency matrix, construct the incidence matrix and also write any three observations on incidence matrix.

(a)
$$A(G) = \begin{bmatrix} 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \end{bmatrix}$$

(b)
$$X(G) = \begin{matrix} & a & b & c & d & e \\ a & \begin{bmatrix} 0 & 1 & 0 & 1 & 0 \end{bmatrix} \\ b & \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \end{bmatrix} \\ c & \begin{bmatrix} 0 & 1 & 0 & 0 & 1 \end{bmatrix} \\ d & \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \end{bmatrix} \\ e & \begin{bmatrix} 0 & 1 & 1 & 1 & 0 \end{bmatrix} \end{matrix}$$

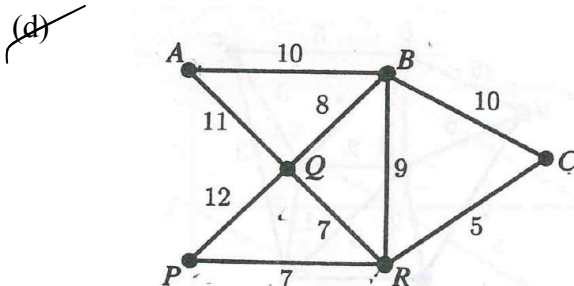
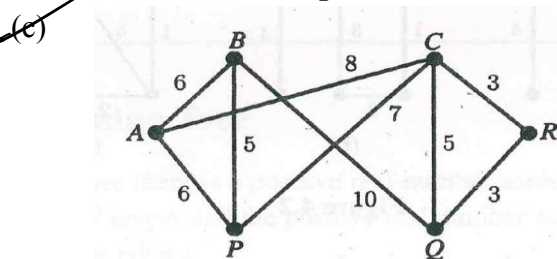
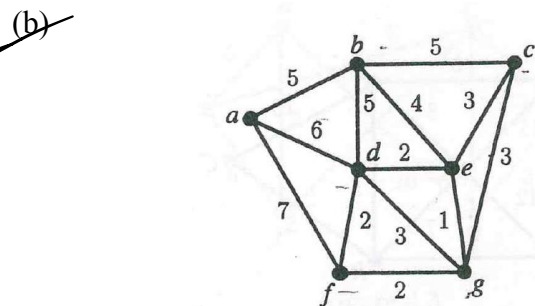
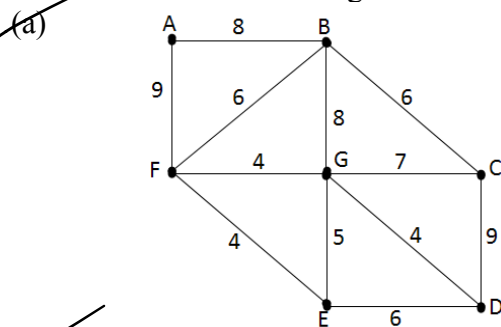
Kruskal's algorithm

30. Eight cities A, B, C, D, E, F, G, H are required to be connected by a new railway network. The possible tracks and the cost involved to lay them (in crores of rupees) are summarized in the table:

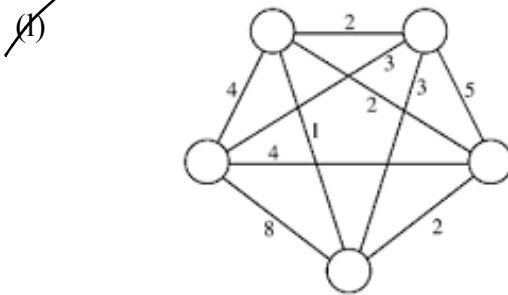
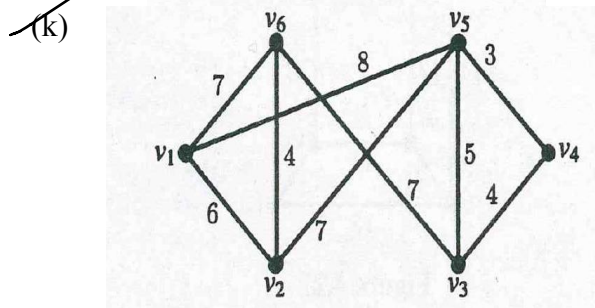
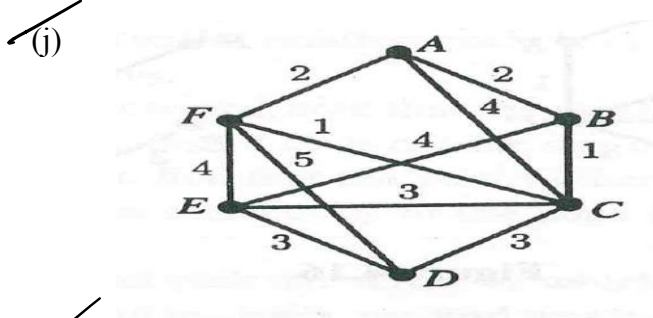
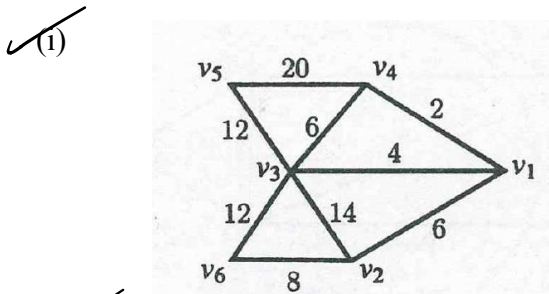
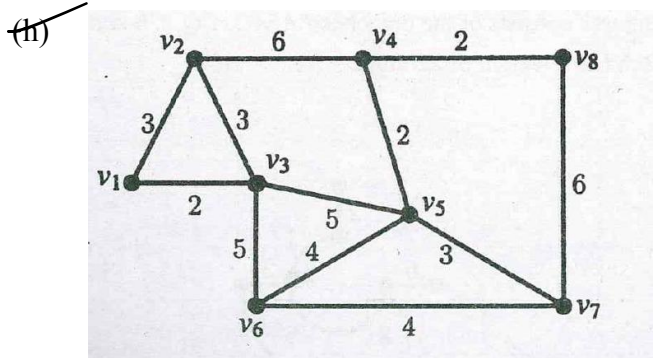
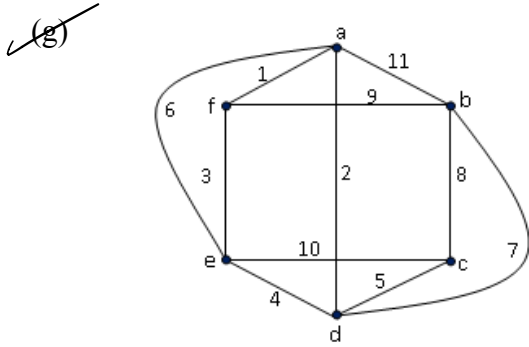
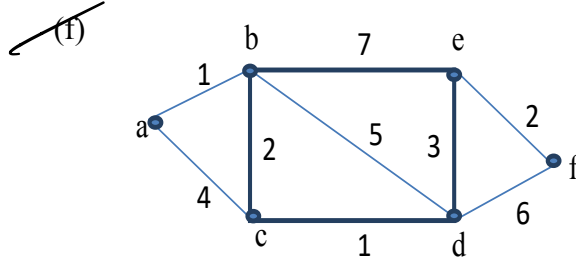
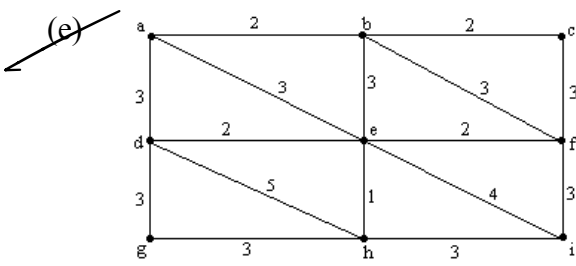
Track between	Cost	Track between	Cost
		C and E	95
A and B	155	D and F	100
A and D	145	E and F	150
A and G	120	F and G	140
B and C	145	F and H	150
C and D	150	G and H	160

Determine a railway network of minimal cost that connects all these cities using Kruskal's algorithm.

31. Apply Kruskal's algorithm to find a minimal spanning tree for the following weighted graphs and hence find its weight.

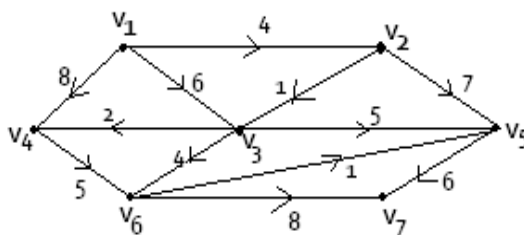


UNIT 1: GRAPH THEORY (23MA3BSSDM)

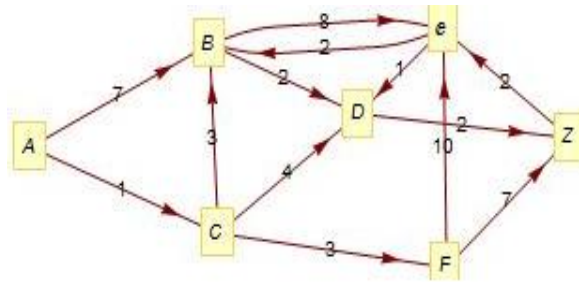


Dijkstra's algorithm

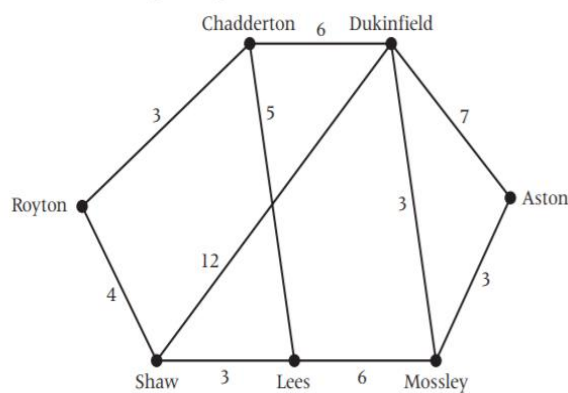
32. Apply Dijkstra's algorithm to find the shortest path and its weight from vertex V_1 to vertex V_5 from the weighted, directed network shown below.



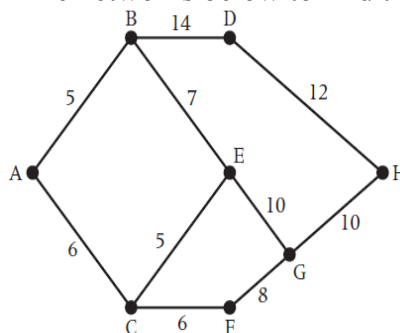
33. Apply Dijkstra's algorithm to find the shortest path and its weight from vertex A to vertex Z from the weighted, directed network



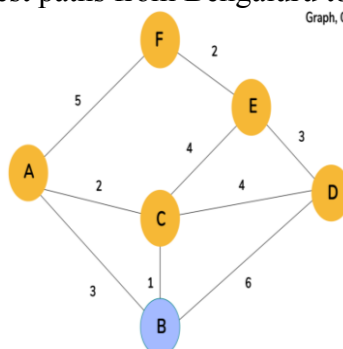
34. The diagram below shows roads connecting villages near to Royton. The numbers on each arc represent the distance, in miles, along each road. Leon lives in Royton and works in Ashton. Use Dijkstra's algorithm to find the minimum distance for Leon's journey to work.



35. Apply Dijkstra's algorithm on the networks below to find the shortest distance from A to H.



36. Consider the map below. The cities have been selected and marked from alphabets A to F and every edge has a cost associated with it. We need to travel from Bengaluru (Vertex B) to all other places. Identify the shortest paths from Bengaluru to other destinations.





STATISTICS AND DISCRETE MATHEMATICS
(Course Code: 23MA3BSSDM)

UNIT-2: PROBABILITY DISTRIBUTIONS

Poisson Distribution

$$X \sim P(\lambda)$$

$$p(x) = \frac{e^{-\lambda} \lambda^x}{x!}; x = 0, 1, 2, 3, \dots$$

1. The traffic police recorded an average of 3 road accidents per week. The number of accidents is distributed according to a Poisson distribution. Calculate the probability in any week of exactly 2 accidents. **Ans: 0.2240**
2. Alpha particles are emitted by radioactive source at the rate of three per every minute on the average. The number of particles is distributed according to the Poisson distribution. Calculate the probability of getting exactly 5 emissions in one minute. **Ans: 0.10082**
3. A distributor of bean seeds determines from extensive tests that 5% of large batch of seeds will not germinate. He sells the seeds in packets of 200 and guarantees 98% germination. Determine the probability that a particular packet will violate the guarantee. **Ans: 0.9995**
4. It is known from past experience that in a certain plant there are on the average 4 industrial accidents per year. Find the probability that in a given year there will be less than 4 accidents. **Ans: 0.4335**
5. In a town, 10 accidents take place in a span of 50 days. Assuming that the number of accidents follows the Poisson distribution, find the probability that there will be 3 or more accidents in a day. **Ans: 0.00115**
6. A book contains 100 misprints distributed randomly throughout its 100 pages. Assuming Poisson distribution, find the probability that a page observed at random contains at least two misprints. **Ans: 0.2642**
7. A switch board can handle only 4 telephone calls per minute. If the incoming calls per minute follow a Poisson distribution with parameter 3, find the probability that the switchboard is over taxed in any one minute. **Ans: 0.1847**
8. A car hire firm has two cars which it hires out day by day. The number of demands for a car on each day is distributed as a Poisson distribution with mean 1.5. Calculate the proportion of days on which (i) there is no demand (ii) demand is refused. **Ans: i) 0.2231 ii) 0.1913**
9. A shop has 4 diesel generator sets which it hires every day. The demand for a generator set on an average is a Poisson variate with value 5/2. Obtain the probability that on a particular day (i) There was no demand (ii) A demand is refused. **Ans: i) 0.08209 ii) 0.10882**
10. A communication channel receives independent pulses at the rate of 12 pulses per micro second. The probability of transmission error is 0.001 for each micro second. Compute the

probabilities of (i) no error during a micro second (ii) one error per micro second (iii) atleast one error per micro second (iv) two errors (v) At most two errors.

Ans: i) 0.9981 ii) 0.01186 iii) 0.1193 iv) 0.000071 v) 0.99999

11. The probability that a news reader commits no mistake in reading the news is e^{-3} . Find the probability that on particular news broadcast he commits (i) only 2 mistakes (ii) more than 3 mistakes (iii) at most 3 mistakes. **Ans: i) 0.2240 ii) 0.3528 iii) 0.6472**
12. In a certain factory turning out razors blades, there is a small chance of 0.002 for any blade to be defective. The blades are supplied in packets of 10. Use Poisson distribution to calculate the approximate number of packets containing (i) no defective (ii) one defective and (iii) two defective blades respectively in a consignment of 10,000 packets. **Ans: i) 9802 ii) 196 iii) 2**
13. The number of accidents in a year to taxi in a city follows a Poisson distribution with mean 3. Out of 1000 taxi drivers find approximately the number of the drivers with (i) no accident in a year (ii) more than 3 accidents in a year. **Ans: i) 50 ii) 350**
14. 2% of the fuses manufactured by a firm are found to be defective. Find the probability that a box containing 200 fuses contains (i) No defective fuses (ii) 3 or more defective fuses. **Ans: i) 0.01831 ii) 0.76189**
15. If the probability of a bad reaction from a certain injection is 0.001, determine the chance that out of 2000 individuals, (i) exactly 3 (ii) more than 2 will suffer a bad reaction. **Ans: i) ii) 0.32**
16. If X follows a Poisson law such that $P(X = 2) = \frac{2}{3} P(X = 1)$, find $P(X=0)$ and $P(X=3)$. **Ans: i) 0.2635 ii) 0.1041**
17. If x is a Poisson variate such that $P(x = 2) = 9 P(x = 4) + 90 P(x = 6)$, compute mean and variance of the Poisson distribution. **Ans: mean = variance = 1**
18. Fit a Poisson distribution for the following data and calculate the theoretical frequencies
- | | | | | | |
|-----|-----|----|----|---|---|
| x | 0 | 1 | 2 | 3 | 4 |
| f | 111 | 63 | 22 | 3 | 1 |
19. Fit a Poisson distribution for the following frequency distribution.
- | | | | | | |
|-----|----|----|----|---|---|
| x | 0 | 1 | 2 | 3 | 4 |
| f | 46 | 38 | 22 | 9 | 1 |
20. Fit a Poisson distribution for the following frequency distribution.
- | | | | | | |
|-----|-----|----|----|---|---|
| x | 0 | 1 | 2 | 3 | 4 |
| f | 122 | 60 | 15 | 2 | 1 |
21. The frequency of accidents per shift in a factory is as shown in the following table:
- | | | | | | |
|-----------------------------|-----|----|----|---|---|
| <i>Accidents per shifts</i> | 0 | 1 | 2 | 3 | 4 |
| <i>Frequency</i> | 180 | 92 | 24 | 3 | 1 |

Calculate the mean number of accidents per shifts and the corresponding Poisson distribution and compare with actual observations.

22. The number of accidents per day (x) as recorded in a textile industry over a period of 400 days is given. Fit a Poisson distribution for the data and calculate the theoretical frequencies.

x	0	1	2	3	4	5
f	173	168	37	18	3	1

Geometric Distribution

$$X \sim G(p)$$

$$p(x) = pq^{x-1}; x = 1, 2, 3, \dots$$

1. The probability that a phone call leads to a sale is 0.4. Calculate the probability that the first sale occurs in the fifth call. **Ans: 0.05184**
2. In a certain town 30% of the people own an iPhone. A researcher asks people at random whether they own an iPhone. The random variable X represents the number of people asked up to and including the first person who owns an iPhone. Determine the chances that:
 - (i) 4 person owns an iPhone
 - (ii) at least 3 people own an iPhone
 - (iii) at most 4 people own an iPhone**Ans: i) 0.1029 ii) 0.49 iii) 0.7599**
3. A representative from the National Football League's Marketing Division randomly selects people on a random street in Kansas City, Kansas until he finds a person who attended the last home football game. Let the probability that he succeeds in finding such a person is 0.20. And, let X denote the number of people he selects until he finds his first success.
 - (i) What is the probability that the marketing representative must select 4 people before he finds one who attended the last home football game?
 - (ii) What is the probability that the marketing representative must select more than 6 people before he finds one who attended the last home football game?
 - (iii) How many people should we expect (that is, what is the average number) the marketing representative needs to select before he finds one who attended the last home football game? And, while we're at it, what is the variance?**Ans: i) 0.1024 ii) 0.2621 iii) 4, 20**
4. An old lawn mower has a 20% chance of starting on a particular pull. Find the probability that it takes (i) exactly 3 pulls to start the mower (ii) 10 or fewer pulls to start the mower
Ans: i) 0.128 ii) 0.8926
5. A person decides to continue placing a bet of Rs.5000 on the number 5 in consecutive spins of a roulette wheel until he wins. On any spin there is a 1 on 50 chances that the ball would land on the number 5.
 - (i) How many spins do you expect until he wins?
 - (ii) What is the amount he is expected to spend until he has his first win?
 - (iii) What are the chances that it takes 5 spins before he wins?
 - (iv) What are the chances that it would take him more than 50 chances to win?**Ans: i) 50 ii) 2,50,000 iii) 0.01845 iv) 0.3642**
6. To finish a board game, a person A needs a sum of 4 with two dice. What is the probability that it takes A under 5 tries to win? How many rolls would you expect A to take until she wins? **Ans: i) 0.294 ii) 12**

7. A practicing shooter scores 93% of his shots during a training session. What are the chances that he would not miss a single shoot till his 20th try? What is the expected number of shots taken before his first miss? **Ans: i) 0.23 ii) 14**

Uniform Distribution (Continuous)

$$X \sim U(a, b)$$

$$f(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

1. If X is uniformly distributed in $-2 \leq X \leq 2$, find (i) $P(X < 1)$ (ii) $P(|X - 1| \geq \frac{1}{2})$. **Ans: i) 0.75 ii) 0.75**
2. If X is uniformly distributed with mean 1 and variance $\frac{4}{3}$, find $P(X < 0)$. **Ans:**
3. A random variable X has a uniform distribution $U[-3, 3]$. Find k if $P(X > 3) = k$. **Ans: 0**
4. A point is chosen at random from the line segment $[0, 2]$. What is the probability that the chosen point lies (i) $1 \leq X \leq \frac{3}{2}$ (ii) $X \leq 1$ (iii) $X \geq 3$ **Ans: i) 0.25 ii) 0.5 iii) 0**
5. If X is uniformly distributed in $[-\alpha, \alpha]$ with $\alpha > 0$ such that $P(X > 1) = \frac{1}{3}$ then determine α . **Ans: 3**
6. A bus travels between two cities A and B which are 100 miles apart. If the bus has a breakdown, the distance X of the point of breakdown from the city A has a uniform distribution $U[0, 100]$. There are service garages in the city A, city B and midway between the two cities such that in case of a breakdown a tow truck is sent from the garage nearest to the point of breakdown.
- What is the probability that the tow truck has to travel more than 10 miles to reach the bus?
 - Would it be more "efficient" if the three service garages were placed at 25, 50 and 75 miles from city A, apart from service garages at city A and city B?
- Ans: i) 0.6 ii) yes**
7. If a conference room cannot be reserved for more than 4 hours, find the probability that a given conference lasts more than 3 hours. **Ans: 0.25**
8. The daily amount of coffee (in liters) dispensed by a machine is uniformly distributed with $a = 7$ and $b = 10$. Determine the probability that the amount of coffee dispensed by the machine will be (i) at most 8.8 liters (ii) more than 7.4 liters but less than 9.5 liters (iii) a least 8.5 liters. **Ans: i) 0.6 ii) 0.7 iii) 0.5**
9. The driving time X from a person's home to the train station is uniformly distributed as $U[10, 50]$. If it takes 2 minutes to board the train, determine the probability that the person catches the 7 am train if he starts at 6:43am from his home. **Ans: 0.125**

- ✓ 10. A bus arrives every 10 minutes at a bus stop. Assuming that the waiting time X for a bus is uniformly distributed, find the probability that the person has to wait for the bus (i) for more than 7 minutes (ii) between 2 and 7 minutes **Ans: i) 0.3 ii) 0.5**
- ✓ 11. The amount charged for a visit to a dental clinic is uniformly distributed from 0 to 1000 (in INR). Given that the amount charged for a visit exceeds Rs 500, calculate the probability that it exceeds Rs. 750. **Ans: 0.5**

Exponential Distribution

$$X \sim E(\alpha)$$

$$f(x) = \begin{cases} \alpha e^{-\alpha x} & x > 0, \alpha > 0 \\ 0 & \text{otherwise} \end{cases}$$

- ✓ 1. In a certain town, the duration of shower is exponentially distributed with mean 5 minutes. What is the probability that the shower will last for (i) less than 10 minutes (ii) 10 minutes or more and (iii) between 10 and 12 minutes. **Ans: i) 0.1353 ii) 0.8647 iii) 0.0446**
- ✓ 2. The length of a telephone conversation in a booth is exponentially distributed and found on an average to be 5 minutes. Find the probability that the random call made from this booth (i) Ends in less than 5 minutes (ii) Between 5 and 10 minutes. **Ans: i) 0.6321 ii) 0.2325**
- ✓ 3. At a certain city bus stop, three buses arrive per hour on an average. Assuming that the time between successive arrivals is exponentially distributed; find the probability that the time between the arrivals of successive buses is (i) less than 10 minutes (ii) at least 30 minutes. **Ans: i) 0.3935 ii) 0.2231**
- ✓ 4. The sales per day in a shop is exponentially distributed with average sale amounting to Rs100/- and net profit is 8%. Find the probability that the net profit exceeds Rs. 30/- on 2 consecutive days. **Ans: 0.00056**
- ✓ 5. The daily turnover in a medical shop is exponentially distributed with Rs.6000 as the average with a net profit of 8%. Find the probability that the net profit exceeds Rs.500 on a randomly chosen day. **Ans: 0.352**
- ✓ 6. Increase in sales per day in a shop is exponentially distributed with mean of Rs.600/-. Sales tax is to be levied at 9%. What is the probability that sales tax will exceed Rs.81 per day? **Ans: 0.2231**
- ✓ 7. The life of a compressor manufactured by a company is known to be 200 months on an average following an exponential distribution. Find the probability that the life of a compressor of that company is (i) Less than 200 months (ii) Between 100 months and 25 years? **Ans: i) 0.6321 ii) 0.3834**
- ✓ 8. The time X (seconds) that it takes a certain online computer terminal (the elapsed time between the end of user's inquiry and the beginning of the system's response to that inquiry) has an exponential distribution with expected time 20 seconds. Compute the probabilities (a) $P(X \leq 30)$ (b) $P(X \geq 20)$ (c) $P(20 \leq X \leq 30)$. **Ans: i) 0.7769 ii) 0.3679 iii) 0.1448**

9. Let the mileage (in thousands of miles) of a particular tyre be a random variable X having the probability density $f(x) = \begin{cases} \frac{1}{20} e^{-x/20} & x > 0 \\ 0 & x \leq 0 \end{cases}$. Find the probability that one of these tyres will last (i) at most 10,000 miles (ii) anywhere from 16,000 to 24,000 miles (iii) at least 30,000 miles (iv) mean (v) variance **Ans: i) 0.3934 ii) 0.148 iii) 0.2231 iv) 20 v) 400**

Normal Distribution/ Gaussian Distribution

$$X \sim N(\mu, \sigma^2)$$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\left(\frac{x-\mu}{\sigma}\right)^2/2}; -\infty < x, \mu < \infty$$

1. In a test on 2000 electric bulbs, it was found that the life of a particular make was normally distributed with an average life of 2040 hours and S.D of 60 hours. estimate the number of bulbs likely to burn for (a) More than 2150 hours (b) Less than 1950 hours (c) More than 1920 hours and but less than 2160 hours.
Ans: a) 67 b) ~~184~~ c) 1909
~~1886~~ = 124
2. In a normal distribution, 31% of the items are under 45 and 8% are over 64. Find the mean and S.D. of the distribution.
Ans: mean=50 S.D=10
3. If the total cholesterol values for a certain population are approximately normally distributed with a mean of 200mg/ml and standard deviation of 20mg/ml. Find the probability that an individual selected at random from this population will have a cholesterol value:
i. Between 180 and 200mg/ml.
ii. Greater than 225mg/ml.
iii. Less than 150mg/ml.
4. In a certain examination, the percentage of candidates passing and getting distinctions were 45 and 9 respectively. Estimate the average marks obtained by the candidates, the minimum pass and distinction marks being 40 and 75 respectively.
Ans: 36.4
5. A manufacturer of air-mail envelopes knows from experience that weight of the envelopes is normally distributed with mean 1.95 gm and S.D. 0.05 gm. About how many envelopes weighting (i) 2 gm or more (ii) 2.05 gm or more can be expected in a given packet of 100 envelopes.
Ans: i) 16 ii) 2
6. The mean height of 500 students is 151 cm and the S.D. is 15 cm. Assuming that the heights are normally distributed, find how many student's heights lie between 120 and 155 cm.
Ans: 294
7. The mean and S.D. of the marks obtained by 1000 students in an examination are respectively 34.4 and 16.5. Assuming the normality of the distribution, find the approximate number of students expected to obtain marks between 30 and 60.
Ans: 543

8. In an examination taken by 500 candidates, the average and S.D. of marks obtained (normally distributed) are 40% and 10%. Find approximately (i) how many will pass, if 50% is fixed as a minimum? (ii) What should be the minimum if 350 candidates are to pass? (iii) How many have scored marks above 60%?

Ans: i) 79 ii) 35% iii) 11

9. The mean inside diameter of a sample of 200 washers produced by a machine is 5.02 mm and the S.D. is 0.05 mm. The purpose for which these washers are intended allows a maximum tolerance in the diameter of 4.96 to 5.08 mm, otherwise the washers are considered defective. Determine the percentage of defective washers produced by the machine, assuming the diameters are normally distributed.

Ans: 23%

10. It is given that the age of thermostats of particular makes follow the normal law with mean 5 years and S.D. 2 years. 1000 units are sold out every month. How many of them will have to be replaced at the end of the second year?

Ans: 67

11. The income of a group of 10,000 persons was found to be normally distributed with mean Rs. 750 p.m. and S.D. of Rs. 50. Show that, of this group, about 95% had income exceeding Rs. 668 and only 5% had income exceeding Rs. 832. Also find the lowest income among the richest 100.

Ans: Rs 866

12. If the top 15% of the students receives A grade and bottom 10% receives F grades in a mathematics examination, determine the (a) minimum marks to get an A grade (b) minimum mark to pass. Assume that the marks are normally distributed with mean 76 and standard deviation 15.

Gamma Distribution

$$X \sim \Gamma(\alpha, \beta)$$

$$f(x) = \begin{cases} \frac{x^{\alpha-1} e^{-x/\beta}}{\beta^\alpha \Gamma(\alpha)} & x > 0 \\ 0 & \text{otherwise} \end{cases}$$

1. The daily consumption of electric power (in million Kw-hours) in a certain city is a random variable X having the probability density

$$f(x) = \begin{cases} \frac{1}{9} x e^{-x/3}, & \text{if } x > 0 \\ 0, & \text{if } x \leq 0 \end{cases}$$

Find the probability that the power supply is adequate on any given day if the capacity of the power plant is 12 million Kw hours. **Ans: 0.9084**

2. The lifetime X (in months) of a computer has a gamma distribution with mean 24 months and standard deviation 12 months. Find the probability that the computer will (i) last between 12 and 24 months (ii) last at most 24 months. **Ans: i) 0.4236 ii) 0.5665**

3. Suppose that the time (in hours) taken by a homeowner to mow his lawns is a random variable having a gamma distribution with parameters $\alpha = 2$ and $\beta = 2$. Find the probability that it takes

(i) at most 1 hour (ii) at least 2 hours (iii) between 0.5 to 1.5 hours to mow the lawn.

Ans: i) 0.5939 ii) 0.09157 iii) 0.5366

- ~~4.~~ The survival time (in weeks) of a male mouse exposed to radiation has a gamma distribution with $\alpha = 8$ and $\beta = 15$. Find the probability that the mouse survives (i) between 60 and 120 weeks (ii) at least 30 weeks. Find the mean and variance of X . **Ans: i) 0.496 ii) 0.9989**
- ~~5.~~ If a random variable has gamma distribution with $\alpha = 2$ and $\beta = 2$. Find (i) mean (ii) standard deviation (iii) the probability that X will take a value less than 4. **Ans i) 4 ii) 2.828 iii) 0.5939**
- ~~6.~~ An actuary models the occurrence of claims from a portfolio of insurance policies. The time until the occurrence of the second claim is modelled by a gamma distribution with mean 10 minutes and variance 50 minutes. Determine the parameters of the distribution. Hence determine the probability that the time until the occurrence of the second claim exceeds 20 minutes. **Ans: 0.0915**
- ~~7.~~ Daily consumption of milk in a town in excess of 20,000 litres is approximately given by Gamma distribution with $\alpha = 3$ and $\beta = 10,000$. The town has a daily stock of 30,000 litres. Find the probability that the stock is insufficient on a given day. **Ans: 0.9198**



STATISTICS AND DISCRETE MATHEMATICS

(Course Code: 22MA3BSSDM)

UNIT-3: JOINT PROBABILITY AND MARKOV CHAINS

Joint Probability: Discrete Case

1. The joint probability distribution of two random variables X and Y is given below:

$X \backslash Y$	2	3	4
1	0.06	0.15	0.09
2	0.14	0.35	0.21

Find the marginal distributions of X and Y . Also verify whether X and Y are stochastically independent.

2. The joint probability distribution of two random variables X and Y is given below:

$X \backslash Y$	-3	2	4
1	0.1	0.2	0.2
3	0.3	0.1	0.1

Find:

- (i) the marginal distributions of X and Y
(ii) $\mu_X, \mu_Y, \sigma_X, \sigma_Y$ and $Cov(X, Y)$
3. The joint probability distribution of two random variables X and Y is given below:

$X \backslash Y$	-4	2	7
1	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{8}$
5	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{8}$

Find: (i) the marginal distributions of X and Y (ii) $E(X)$ and $E(Y)$ (iii) σ_X, σ_Y (iv) $Cov(X, Y)$ (v) $\rho(X, Y)$

4. The joint probability distribution of two random variables X and Y is given below:

$X \backslash Y$	1	3	9
2	$\frac{1}{8}$	$\frac{1}{24}$	$\frac{1}{12}$
4	$\frac{1}{4}$	$\frac{1}{4}$	0
6	$\frac{1}{8}$	$\frac{1}{24}$	$\frac{1}{12}$

Find the marginal distributions of X and Y . Also evaluate $Cov(X, Y)$.

5. The joint probability function of two random variables X and Y is given by $f(x, y) = c(2x + y)$ where x and y can assume all integral values such that $0 \leq x \leq 2, 0 \leq y \leq 3$ and $f(x, y) = 0$, otherwise. Find:

- (i) the value of the constant c
 - (ii) $P(X \geq 1, Y < 2)$
 - (iii) Marginal Probability distribution of X and Y .
 - (iv) Check whether X and Y are independent.
6. A coin is tossed three times. Let X denote 0 or 1 according as tail or head occurs on the first toss. Let Y denote the total number of tails which occur. Determine:
 - (i) the marginal distributions of X and Y
 - (ii) the joint distributions of X and Y . Also find the expected values of $X + Y$ and XY .
 7. Two flashcards are selected at random from a box which contains five cards numbered 1, 1, 2, 2 and 3. Find the joint distributions of X and Y where X denotes the sum of two numbers and Y denote the maximum of two numbers drawn. Also determine $Cov(X, Y)$.
 8. X and Y are independent random variables. X takes values 2, 5, 7 with probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{4}$ respectively. Y takes values 3, 4, 5 with probabilities $\frac{1}{3}, \frac{1}{3}, \frac{1}{3}$.
 - (i) Find the joint probability distribution of X and Y
 - (ii) Show that the covariance of X and Y is equal to zero
 - (iii) Find the probability distribution of $Z = X + Y$
 9. Two fruits are selected at random from a bag containing 3 Apples, 2 Oranges and 4 Mangoes. If X and Y are respectively, the number of Apples and the number of Oranges included among the two fruits drawn from the bag, find the probability associated with all possible pair of values (x, y) . Also find the correlation between the variables X and Y .
 10. Two marbles are selected at random from a box containing 3 Blue, 2 Red and 3 Green marbles. If X and Y are respectively, the number of Blue marbles and the number of Red marbles selected, find:
 - (i) the probability associated with all possible pair of values
 - (ii) Show that the random variables X and Y are not stochastically independent.

Joint Probability: Continuous Case

1. Find the constant k so that

$$f(x, y) = \begin{cases} k(x+1)e^{-y}, & 0 < x < 1, y > 0 \\ 0 & \text{elsewhere} \end{cases}$$

is a joint probability density function. Are x and y independent?

2. Suppose the joint p.d.f of (x, y) is given by

$$p(x, y) = \begin{cases} e^{-y}, & x > 0, y > x \\ 0 & \text{elsewhere} \end{cases}$$

Find: (i) the marginal density function of x (ii) the marginal density function of y
Check whether x and y are independent.

3. Let x and y be random variables having the joint density function

$$f(x, y) = \begin{cases} 4xy, & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the marginal density functions of x and y . Verify that $E(x + y) = E(x) + E(y)$ and $E(xy) = E(x)E(y)$

4. The joint probability density function of two random variables x and y is given by

$$p(x, y) = \begin{cases} 2, & 0 < x < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Find the marginal density functions of x and y . Also find the covariance between x and y .

5. The joint density function of two random variables x and y is given by

$$f(x, y) = \begin{cases} kxy, & 0 < x < 4, 1 < y < 5 \\ 0 & \text{elsewhere} \end{cases}$$

Find: (i) the value of k (ii) $E(x), E(y), E(xy), E(2x + 3y)$

6. If the joint probability function for (x, y) is

$$f(x, y) = \begin{cases} c(x^2 + y^2), & 0 \leq x \leq 1, 0 \leq y \leq 1, c \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Determine the following: (i) the value of the constant c (ii) marginal density functions of x and y (iii) $P\left(x < \frac{1}{2}, y > \frac{1}{2}\right)$ (iv) $P\left(\frac{1}{4} < x < \frac{3}{4}\right)$ (v) $P\left(y < \frac{1}{2}\right)$

7. The joint probability function for (x, y) is given by

$$f(x, y) = \begin{cases} cxy, & 0 \leq x \leq 2, 0 \leq y \leq x \\ 0 & \text{otherwise} \end{cases}$$

Determine c and hence evaluate $P\left(\frac{1}{2} < x < 1\right)$

8. Verify that $f(x, y) = \begin{cases} e^{-(x+y)}, & x \geq 0, y \geq 0 \\ 0 & \text{otherwise} \end{cases}$

is a density function of a joint probability distribution. Hence evaluate:

- (i) $P\left(\frac{1}{2} < x < 2, 0 < y < 4\right)$ (ii) $P(x < 1)$ (iii) $P(x \leq y)$ (iv) $P(x > y)$
(v) $P(x + y \leq 1)$

9. Suppose that the joint density function of two random variables x and y is

$$f(x, y) = \begin{cases} x^2 + \frac{1}{3}xy, & 0 < x < 1, 0 < y < 2 \\ 0 & \text{elsewhere} \end{cases}$$

Evaluate the following:

- (i) $P\left(x \geq \frac{1}{2}\right)$ (ii) $P(x + y \geq 1)$ (iii) $P(Y \leq x)$

10. For the distribution given by the density function

$$f(x, y) = \begin{cases} \frac{1}{96}xy, & 0 < x < 4, 1 < y < 5 \\ 0 & \text{otherwise} \end{cases}$$

Evaluate: (i) $P(1 < x < 2, 2 < y < 3)$ (ii) $P(x \geq 3, y \leq 2)$ (iii) $P(x + y \leq 3)$ (iv) $P(x + y \leq 3)$

1. Alice and Bob meet for lunch every day at a random time between noon and 1 P.M. Bob always arrives after Alice, but otherwise, they are equally likely to arrive at any time. The joint p.d.f. of X , the time Alice arrives (in minutes after 12 P.M.), and Y , the time Bob arrives (in minutes after 12 P.M.), is given

$$f(x, y) = \begin{cases} c & 0 \leq x < y \leq 60 \\ 0 & \text{otherwise} \end{cases}$$

where c is a constant.

- Sketch a birds-eye view of the joint p.d.f. (Because the p.d.f. is constant, you should be able to calculate all probabilities using geometry, rather than integration).
 - Determine the value of c that makes this a valid joint p.d.f.
 - Calculate the probability that Bob arrives more than 25% later than Alice. That is, what is $P(Y > 1.25X)$? (Sketch a picture of this region.)
 - Suppose X and Y are continuous random variables with joint p.d.f. $f(x, y)$. What is $P(X=Y)$ and why?
2. An ecologist selects a point inside a circular sampling region according to a uniform distribution. Let X be the x -coordinate of the point selected and Y be the y -coordinate of the point selected. If the circle is centered at $(0,0)$ and has radius r , then the joint pdf of X and Y is

$$f(x, y) = \begin{cases} c^2 & x^2 + y^2 \leq r^2 \\ 0 & \text{otherwise} \end{cases}$$

- Determine the value of c that makes this a valid joint p.d.f.
 - What is the probability that the selected point is within $r/2$ of the center of the circular region? (Hint: Use geometry.)
 - What is the probability that *both* X and Y differ from 0 by at most $r/2$?
3. A company produces cans of mixed nuts containing almonds, cashews, and peanuts. Each can is exactly 1 lb, but the amount of each type of nut is random. The joint p.d.f. of X , the amount of almonds, and Y , the amount of cashews, is $f(x, y) = \begin{cases} 24xy & 0 \leq x \leq 1, 0 \leq y \leq 1, x + y \leq 1, \\ 0 & \text{otherwise} \end{cases}$

Show that the probability there are more almonds than cashews is 0.50.

Markov Chain

1. Define (i) Stochastic matrix (ii) Absorbing State (iii) Fixed Probability Vector.
2. Verify that the matrix $A = \begin{pmatrix} 0 & 0 & 1 \\ 0.5 & 0.25 & 0.25 \\ 0 & 1 & 0 \end{pmatrix}$ is a regular stochastic matrix.
3. Verify that the matrix $A = \begin{pmatrix} 0 & 1 \\ 0.3 & 0.7 \end{pmatrix}$ is a regular stochastic matrix.

4. Prove that the Markov Chain whose transition probability matrix is $P = \begin{bmatrix} 0 & 2/3 & 1/3 \\ 1/2 & 0 & 1/2 \\ 1/2 & 1/2 & 0 \end{bmatrix}$ is irreducible.
5. Find the fixed probability vector for the transition matrix $P = \begin{bmatrix} 0.1 & 0.3 & 0.6 \\ 0 & 0.4 & 0.6 \\ 0 & 0.2 & 0.8 \end{bmatrix}$.
6. Find the unique fixed probability vector for the regular stochastic matrix $A = \begin{pmatrix} 0 & 1 & 0 \\ 1/6 & 1/2 & 1/3 \\ 0 & 2/3 & 1/3 \end{pmatrix}$.
7. Find the unique fixed probability vector for the regular stochastic matrix $A = \begin{pmatrix} 0.75 & 0.25 \\ 0.5 & 0.5 \end{pmatrix}$.
8. Find the unique fixed probability vector for the regular stochastic matrix $A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/4 & 1/4 \end{pmatrix}$.
9. A student's study habits are as follows. If he studies one night, he is 70% sure not to study the next night. On the other hand, if he does not study one night, he is 60% sure not to study the next night. In a long run how often does he study?
10. Consider a game of "ladder climbing". There are 5 levels in the game, level 1 is the lowest (bottom) and level 5 is the highest (top). A player starts at the bottom. Each time, a fair coin is tossed. If it turns up heads, the player moves up one rung. If tails, the player moves down to the very bottom. Once at the top level, the player moves to the very bottom if a tail turns up, and stays at the top if head turns up.
- Find the transition probability matrix.
 - Find the two-step transition probability matrix.
 - Find the steady-state distribution of the Markov chain.
11. The pattern of sunny and rainy days on the planet Rainbow is a homogeneous Markov chain with two states. Every sunny day is followed by another sunny day with probability 0.8. Every rainy day is followed by another rainy day with probability 0.6.
- If today is sunny on Rainbow, what is the chance of rain the day after tomorrow?
 - Compute the probability that April 1 next year is rainy on Rainbow.
12. A computer device can be either in a busy mode (state 1) processing a task, or in an idle mode (state 2), when there are no tasks to process. Being in a busy mode, it can finish a task and enter an idle mode any minute with the probability 0.2. Thus, with the probability 0.8 it stays another minute in a busy mode. Being in an idle mode, it receives a new task any minute with the probability 0.1 and enters a busy mode. Thus, it stays another minute in an idle mode with the probability 0.9. The initial state is idle. Let X_n be the state of the device after n minutes.
- Find the distribution of X_2 .
 - Find the steady-state distribution of X_n .

13. In a certain city, the weather on a day is reported as sunny, cloudy or rainy. If a day is sunny, the probability that the next day is sunny is 70%, cloudy is 20% and rainy is 10%. If a day is cloudy, the probability that the next day is sunny is 30%, cloudy is 20% and rainy is 50%. If a day is rainy, the probability that the next day is sunny is 30%, cloudy is 30% and rainy is 40%. If a Sunday is sunny, find the probability that the Wednesday is rainy.
14. A housewife buys 3 kinds of cereals A, B, and C. She never buys the same cereal in successive weeks. If she buys cereal A, the next week she buys cereal B. However, if she buys cereal B or C, the next week she is three times as likely to buy cereal A as the other cereal. In the long run how often she buys each of the three cereals?
15. A salesman's territory consists of 3 cities A, B and C. He never sells in the same city on successive days. If he sells in city A, then the next day he sells in city B. However, if he sells in either B or C, then the next day he is twice as likely to sell in city A as in other city. In the long run, how often does he sell in each of the cities?
16. A housewife buys 3 brands of soaps: A, B, C. She never buys the same brand on successive weeks. If she buys brand A in a week, she buys brand B in the next week. If she buys the brand other than A in a week, then in the next week she is three times as likely to buy the brand A as the other brand. Supposing that she has bought brand B in the first week, find the probability of her buying each of the three brands in the fourth week.
17. Each year a man trades his car for a new car in 3 brands of the popular company Maruti Udyog Limited. If he has "Standard", he trades it for "Zen". If he has a "Zen", he trades it for "Esteem". If he has "Esteem", he is just as likely to trade it for a new "Esteem" or for a "Zen" or a "Standard". In 2010 he bought his first car which was "Esteem". Find the probability that he has on 2012 "Esteem" and on 2013 "Zen".
18. Three boys A, B, C are throwing ball to each other. A always throws the ball to B and B always throws the ball to C. C is just as likely to throw the ball to B as to A. If C was the first person to throw the ball. Find the probability after three throws that
(i) A has the ball (ii) B has the ball (iii) C has the ball
19. A habitual gambler is a member of two clubs A and B. He visits either of the clubs everyday for playing cards. He never visits club A on two consecutive days. But, if he visits club B on a particular day, then the next day he is as likely to visit club B or club A.
(i) Find the transition matrix of this Markov chain.
(ii) Show that the matrix is a regular stochastic matrix.
(iii) Find the unique fixed probability vector.
(iv) If the person had visited club B on Monday, find the probability that he visits club A on Thursday.
20. A gambler's luck follows a pattern. If he wins a game, the probability of winning the next game is 0.6. However, if he loses a game, the probability of losing the next game is 0.7. There is an even chance of gambler winning the first game. If so
i. What is the probability of he winning the second game?
ii. What is the probability of he winning the third game?
iii. In the long run, how often he will win?

B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19
(Autonomous Institute, Affiliated to VTU)
DEPARTMENT OF MATHEMATICS
THIRD SEMESTER B.E COURSE(CSE/ISE/CSE-AIDS/CS-IOT/CSE-DS)

Course Title: Statistics and Discrete Mathematics
Course Code: 23MA3BSSDM

UNIT 4: STATISTICAL INFERENCE

I. Test of significance for single mean [large sample: $n \geq 30$]

Test statistic	Null Hypothesis	Alternative Hypothesis	Accept H_o
$Z = \frac{\bar{x} - \mu_o}{\sigma / \sqrt{n}}$	$H_0 : \mu = \mu_o$	$H_1 : \mu \neq \mu_o$	$ z < z_{\alpha/2}$
		$H_1 : \mu < \mu_o$	$z > z_{\alpha}$
		$H_1 : \mu > \mu_o$	$z < z_{\alpha}$

- The length of life X of certain computers is approximately normally distributed with mean 800 hours and S.D 40 hours. If a random sample of 30 computers has an average life of 788 hours, test the null hypothesis that $\mu=800$ hours at (a) 5% (b) 1% (c) 10% (d) 15% level of significance
- Mice with an average lifespan of 32 months will live upto 40 months when fed by a certain nutritious food. If 64 mice fed on this diet have an average lifespan of 38 months and standard deviation of 5.8 months, is there any reason to believe that average lifespan is less than 40 months.
- A machine runs on an average of 125 hours/year. A random sample of 49 machines has an annual average use of 126.9 hours with standard deviation 8.4 hours. Does this suggest to believe that machines are used on the average more than 125 hours annually at 0.05 level of significance?
- A company claims that the mean thermal efficiency of diesel engines produced by them is 32.3%. To test this claim, a random sample of 40 engines were examined which showed the mean thermal efficiency of 31.4% and S.D of 1.6%. Can the claim be accepted or not, at 0.01 L.O.S.?
- It has previously been recorded that the average depth of ocean at a particular region is 67.4 fathoms. Is there reason to believe this at 0.01 L.O.S. if the readings at 40 random locations in that particular region showed a mean of 69.3 with S.D of 5.4 fathoms?
- A company producing computers states that the mean lifetime of its computers is 1600 hours. Test this claim at 0.01 L.O.S. against the A.H.: $\mu < 1600$ hours if 100 computers produced by this company has mean lifetime of 1570 hours with S.D. of 120 hours.
- A manufacturer of tyres guarantees that the average lifetime of its tyres is more than 28000 miles. If 40 tyres of this company tested, yields a mean lifetime of 27463 miles with S.D. of 1348 miles, can the guarantee be accepted at 0.01 L.O.S.?

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DEPARTMENT OF MATHEMATICS

II. Test of significance for difference between two means [large sample: $n \geq 30$]

Test statistic	Null Hypothesis	Alternative Hypothesis	Accept H_o
$Z = \frac{(\bar{x}_1 - \bar{x}_2) - \delta}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$	$H_0 : \mu_1 - \mu_2 = \delta$	$H_1 : \mu_1 - \mu_2 \neq \delta$	$ z < z_{\alpha/2}$
		$H_1 : \mu_1 - \mu_2 > \delta$	$z < z_\alpha$
		$H_1 : \mu_1 - \mu_2 < \delta$	$z > z_\alpha$

1. In a random sample of 100 tube lights produced by company A, the mean lifetime (mlt) of tube light is 1190 hours with standard deviation of 90 hours. Also in a random sample of 75 tube lights from company B the mean lifetime is 1230 hours with standard deviation of 120 hours. Is there a difference between the mean lifetimes of the two brands of tube lights at a significance level of (a) 0.05 (b) 0.01?
2. To test the effects a new pesticide on rice production, a farm land was divided into 60 units of equal areas, all portions having identical qualities as to soil, exposure to sunlight etc. The new pesticide is applied to 30 units while old pesticide to the remaining 30. Is there reason to believe that the new pesticide is better than the old pesticide if the mean number of kgs of rice harvested/units using new pesticide (N.P.) is 496.31 with S.D. of 17.18 kgs. Test at a level of significance (a) 0.05 (b) 0.01?
3. A random sample of 40 'geysers' produced by company A have a mean lifetime (mlt) of 647 hours of continuous use with a S.D. of 27 hours, while a sample 40 produced by another company B have mlt of 638 hours with S.D. 31 hours. Does this substantiate the claim of company A that their 'geysers' are superior to those produced by company B at (a) 0.05 (b) 0.01 L.O.S.
4. Test the N.H.: $\mu_A - \mu_B = 0$ against the A.H.: $\mu_A - \mu_B \neq 0$ at 0.01 L.O.S. for the following data

	Sample Size	Mts (kgs)	S.D. (kgs)
Type A	40	247.3	15.2
Type B	30	254.1	18.7

5. If a random sample data show that 42 men earn on the average $\bar{x}_1 = 744.85$ with S.D. $s_1 = 397.7$ while 32 women earn on the average $\bar{x}_2 = 516.78$ with S.D. $s_2 = 162.523$, test at 0.05 level of significance whether the average income for men and women is same or not.

B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19*(Autonomous Institute, Affiliated to VTU)***DEPARTMENT OF MATHEMATICS****III. Small Sample Test Concerning Single Mean: t-Distribution**

Test statistic	Null Hypothesis	Alternative Hypothesis	Accept H_o
$t = \frac{\bar{x} - \mu_o}{S/\sqrt{n}} \text{ here } S^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$ OR $t = \frac{\bar{x} - \mu_o}{s/\sqrt{n-1}} \text{ here } s^2 = \frac{\sum (x_i - \bar{x})^2}{n}$	$H_0 : \mu = \mu_o$	$H_1 : \mu \neq \mu_o$	$ t < t_{\alpha/2, n-1}$
		$H_1 : \mu < \mu_o$	$t > t_\alpha$
		$H_1 : \mu > \mu_o$	$t < t_\alpha$

1. An ambulance service company claims that on an average it takes 20 minutes between a call for an ambulance and the patient's arrival at the hospital. If in 6 calls the time taken (between a call and arrival at hospital) are 27, 18, 26, 15, 20 and 32. Can the company's claim be accepted? **(5%) (1%)**
2. Mean lifetime of computers manufactured by a company is 1120 hours with standard deviation of 125 hours. (a) Test the hypothesis that mean lifetime of computers has not changed if a sample of 8 computers has a mean life time of 1070 hours (b) Is there decrease in mlt? Use (i) 5% (ii) 1% L.O.S.
3. An auditor claims that he takes on an average 10.5 days to file income tax returns (I.T. returns). Can this claim be accepted if a random sample shows that he took 13, 19, 15, 10, 12, 11, 14, 18 days to file I.T. returns? Use (a) 0.01 (b) 0.05 L.O.S.
4. If 5 pieces of certain ribbon selected at random have mean breaking strength of 169.5 pounds with S.D. of 5.7, do they confirm to the specification mean breaking strength of 180 pounds.
5. In a random sample of 10 bolts produced by a machine the mean length of bolt is 0.53 mm and S.D 0.03 mm. Can we claim from this that the machine is in proper working order if in the past it produced bolts of length 0.50 mm? Use (a) 0.05 (b) 0.01 L.O.S.

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IV. Small Sample Test Concerning Difference Between Two Means: t-Distribution

Test statistic	Null Hypothesis	Alternative Hypothesis	Accept H_0
$t = \frac{(\bar{x}_1 - \bar{x}_2) - \delta}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$	$H_0 : \mu_1 - \mu_2 = \delta$	$H_1 : \mu_1 - \mu_2 \neq \delta$	$ t < t_{\alpha/2, n_1+n_2-2}$
(i) $S^2 = \frac{\sum (x_{1i} - \bar{x}_1)^2 + \sum (x_{2i} - \bar{x}_2)^2}{n_1 + n_2 - 2}$ OR (ii) $S^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}$ if $s_1^2 = \frac{\sum (x_{1i} - \bar{x}_1)^2}{n_1}$ and $s_2^2 = \frac{\sum (x_{2i} - \bar{x}_2)^2}{n_2}$		$H_1 : \mu_1 - \mu_2 > \delta$	$t < t_\alpha$
OR (iii) $S^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$ if $S_1^2 = \frac{\sum (x_{1i} - \bar{x}_1)^2}{n_1 - 1}$ and $S_2^2 = \frac{\sum (x_{2i} - \bar{x}_2)^2}{n_2 - 1}$		$H_1 : \mu_1 - \mu_2 < \delta$	$t > t_\alpha$

1. In a mathematics examination 9 students of class A and 6 students of class B obtained the following marks. Test at 1% L.O.S. whether the performance in mathematics is same or not for the two classes A and B. assume that the samples are drawn from normal populations having same variance.

A:	44	71	63	59	68	46	69	54	48
B:	52	70	41	62	36	50	-	-	-

2. Out of random sample of 9 mice, suffering with a disease, 5 mice were treated with a new serum while the remaining were not treated. From the time commencement of experiment, the following are the survival times:

Treatment	2.1	5.3	1.4	4.6	0.9
NoTreatment	1.9	0.5	2.8	3.1	-

Test whether the serum treatment is effective in curing the disease at 5% L.O.S., assuming that the two distributions are normally distributed with equal variances.

3. Random samples of specimens of coal from two mines A and B are drawn and their heat producing capacity (in millions of calories/ton) were measured yielding the following results:

Mine A:	8350	8070	8340	8130	8260	-
Mine B:	7900	8140	7920	7840	7890	7950

Is there significant difference between the means of these two samples at 1% L.O.S.?

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4. To determine whether vegetarian and non-vegetarian diets effects significantly on increase in weight a study was conducted yielding the following data of gain in weight.

Veg	34	24	14	32	25	32	30	24	30	31	35	25	-	-	-
Non-Veg	22	10	47	31	44	34	22	40	30	32	35	18	21	35	29

Can we claim that the two diets differ pertaining to weight gain, assuming that samples are drawn from normal populations with same variance?

5. A group of 10 rats fed on a diet A and another group of 8 rats fed on a different diet B recorded the following increase in weights.

Diet A	5	6	8	1	12	4	3	9	6	10
Diet B	2	3	6	8	10	1	2	8	-	-

Does it show the superiority of diet A over that of B?

6. A group of 5 patients treated with medicine “A” weigh 42, 39, 48, 60, and 41 kgs. A second group of 7 patients from the same hospital treated with medicine “B” weigh 38, 42, 56, 64, 68, 69 and 62 kgs. Do you agree with the claim that the medicine “B” increases the weight significantly?

B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19*(Autonomous Institute, Affiliated to VTU)***DEPARTMENT OF MATHEMATICS****V. Paired Sample t-Test**

Test statistic	Null Hypothesis	Alternative Hypothesis	Accept H_o
$t = \frac{\bar{d} - \mu_d}{S_d / \sqrt{n}} \text{ here } S_d^2 = \frac{\sum (d_i - \bar{d})^2}{n-1}$ OR $t = \frac{\bar{d} - \mu_d}{S_d / \sqrt{n-1}} \text{ here } S_d^2 = \frac{\sum (d_i - \bar{d})^2}{n}$	$H_0 : \mu = \mu_d$	$H_1 : \mu \neq \mu_d$	$ t < t_{\alpha/2, n-1}$
		$H_1 : \mu < \mu_d$	$t > t_\alpha$
		$H_1 : \mu > \mu_d$	$t < t_\alpha$

1. Use paired sample test at 0.05 L.O.S. to test from the following data whether the differences of means of the weights obtained by two different scales (weighting machines) is significant.

Scale: I	11.23	14.36	8.33	10.50	23.42	9.15	13.47	6.47	12.40	19.38
Scale: II	11.27	14.41	8.35	10.52	23.41	9.17	13.52	6.46	12.45	19.35

2. The average weekly losses of man-hours due to strikes in an institute before and after a disciplinary program was implemented are as follows: Is there reason to believe that the disciplinary program is effective at 5% L.O.S.?

Before	45	73	46	124	33	57	83	34	26	17
After	36	60	44	119	35	51	77	29	24	11

3. The pulsality index (P.I.) of 11 patients before and after contracting a disease are given below. Test at 0.05 L.O.S. whether there is a significant increase of the mean of P.I. values.

Before	0.4	0.45	0.44	0.54	0.48	0.62	0.48	0.60	0.45	0.46	0.35
After	0.5	0.60	0.57	0.65	0.63	0.78	0.63	0.80	0.69	0.62	0.68

4. The blood pressure (B.P.) of 5 women before and after intake of a certain drug are given below:

Before	110	120	125	132	125
After	120	118	125	136	121

Test at 1% L.O.S. whether there is significant change in B.P.

5. Marks obtained in mathematics by 11 students before and after intensive coaching are given below:

Before	24	17	18	20	19	23	16	18	21	20	19
After	24	20	22	20	17	24	20	20	18	19	22

Test at 5% L.O.S. whether the intensive coaching is useful?

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VI. Ratio of Variances: F-Distributions

Test statistic	Null Hypothesis	Alternative Hypothesis	Accept H_o
$F = \frac{S_1^2}{S_2^2}$ if $S_1^2 > S_2^2$ OR $F = \frac{S_2^2}{S_1^2}$ if $S_2^2 > S_1^2$ here $S_1^2 = \frac{n_1 s_1^2}{n_1 - 1} = \frac{\sum (x_{1i} - \bar{x}_1)^2}{n_1 - 1};$ $S_2^2 = \frac{n_2 s_2^2}{n_2 - 1} = \frac{\sum (x_{2i} - \bar{x}_2)^2}{n_2 - 1}$	$H_0 : \sigma_1^2 = \sigma_2^2$	$H_1 : \sigma_1^2 \neq \sigma_2^2$	$F < F_{n_1-1, n_2-1, \alpha}$ OR $F < F_{n_2-1, n_1-1, \alpha}$

1. The household net expenditure on health care in south and north India, in two samples of households, expressed as percentage of total income is shown the following table:

South	15	8	3.8	6.4	27.4	19	35.3	13.6	
North	18.8	23.1	10.3	8	18	10.2	15.2	19	20.2

Test the equality of variances of household's net expenditure on health care in south and north India.

2. Can we conclude that the two population variances are equal for the following data of post graduates passed out from a state and private university.

State:	8350	8260	8130	8340	8070	
Private:	7890	8140	7900	7950	7840	7920

3. Is there reason to believe that the life expected in south and north India is same of not from the following data

South:	34	39.2	46.1	48.7	49.4	45.9	55.3	42.7	43.7		
North:	49.7	55.4	57	54.2	50.4	44.2	53.4	57.5	61.9	56.6	58.2

B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19*(Autonomous Institute, Affiliated to VTU)***DEPARTMENT OF MATHEMATICS****VII. Chi-Square Distribution: Goodness of Fit**

Test statistic	Null Hypothesis	Alternative Hypothesis	Accept H_o
$\chi^2 = \frac{\sum_{i=1}^n (o_i - e_i)^2}{e_i}$	H_0 : There is no significant difference between experimental and theoretical values	H_1 : There is significant difference between experimental and theoretical values	$\chi^2 < \chi^2_{n-k, \alpha}$

- 1) A sample analysis of examination results of 500 students was made. It was found that 220 students had failed, 170 had secured third class, 90 had secured second class and 20 had secured first class. Do these figures support the general examination result which is in the ratio 4:3:2:1 for the respective categories?
- 2) In a Mendelian experiment on breeding, four types of plants are expected to occur in the proportion 9:3:3:1. The observed frequencies are 891 round and yellow, 316 wrinkled and yellow, 290 round and green, and 119 wrinkled and green. Find the Chi-square value and examine the correspondence between the theory and the experiment.
- 3) A machine is supposed to mix peanuts, hazelnuts, cashews and pecans in the ratio 5:2:2:1. A can containing 500 of these mixed nuts was found to have 269 peanuts, 112 hazel nuts, 74 cashews and 45 pecans. Can we conclude that the machine is mixing the nuts in the stated ratio?
- 4) It is believed that the proportion of people with A, B, O and AB blood types in a population are respectively 0.4, 0.2, 0.3 and 0.1. When 400 randomly picked people were examined, the number of persons with these types was observed to be 148, 96, 106 and 50 respectively. Test the hypothesis that these data bear out the stated belief.
- 5) Genetic theory states that children having one parent of blood type M and other blood type N will always be one of three types M, MN, N and that the proportions of these types will on average be 1: 2: 1. A report states that out of 300 children having one parent M and one N parent 30% were found to be of type M, 45% of type MN and remainder of type N. Find the Chi-square value and examine the correspondence between the theory and the experiment.
- 6) Among 64 off-springs of a certain cross between pigs, 34 were red, 10 were black and 20 were white. According to a genetic model, these numbers should be in proportions 9:3:4. Is the data consistent with the model at 5% level?
- 7) In 1000 extensive sets of trials for an event of small probability, the frequency of the number x of seeds proved to be

x	0	1	2	3	4	5	6	7
f	304	366	210	80	28	9	2	1

Fit a Poisson distribution to the data and test the goodness of fit.

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- 8) The number of cars passing a given point in 100 five second interval was observed as follows

No of Cars	0	1	2	3	4	5
No of intervals	40	30	14	6	5	5

Fit a Poisson distribution and test for its goodness of fit.

- 9) Test for goodness of fit of a Poisson distribution at 5% L.O.S. to the following frequency distribution:

No. of patients arriving/hour: (x)	0	1	2	3	4	5	6	7	8
Frequency	52	151	130	102	45	12	5	1	2

- 10) It is desired to test whether the number of gamma rays emitted per second by a certain radioactive substance is a random variable having the Poisson distribution with mean 2.4. Use the following data obtained for 300 one-second intervals to test this null hypothesis.

Number of gamma rays	0	1	2	3	4	5	6	7 or more
Frequency	18	48	66	74	44	35	10	5



STATISTICS AND DISCRETE MATHEMATICS

(Course Code: 23MA3BSSDM)

UNIT-5: COMBINATORICS

I. Binomial Theorem and multinomial theorem:

1. What is the coefficient of $x^{12}y^{13}$ in the expansion of $(2x-3y)^{25}$.
2. Find the coefficient of
 - (i) x^9y^3 in the expansion of $(2x-3y)^{12}$
 - (ii) x^0 in the expansion of $\left(3x^2 - \frac{2}{x}\right)^{15}$
 - (iii) x^{12} in the expansion of $x^3(1-2x)^{10}$
3. Determine the coefficient of
 - (i) xyz^2 in the expansion of $(2x-y-z)^4$
 - (ii) $x^2y^2z^3$ in the expansion of $(3x-2y-4z)^7$
 - (iii) $a^2b^3c^2d^5$ in the expansion of $(a+2b-3c+2d+5)^{16}$

II. The Principle of Inclusion Exclusion and Generalizations of The Principles:

1. Out of 1200 students at a college, 582 took economics, 627 took English, 543 to mathematics 217 took both economics and English, 307 took both economics and mathematics, 250 took both mathematics and English and 212 took all three courses. How many took none of the three?
2. Out of 30 students in a hostel, 15 study History, 8 study Economics and 6 study Geography. It is known that 3 students study all these subjects. Show that 7 or more students' study none of these subjects.
3. Among the first 500 positive integers, determine the integers which are not divisible by 2, nor by 3 nor by 5.
4. Among 100 students, 32 study mathematics, 20 study physics, 45 study biology, 15 study mathematics and biology, 7 study mathematics and physics, 10 study biology and physics and 30 do not study any of the subjects. Find the number of students studying all three subjects?
5. How many integers between 1 and 300 (inclusive) are
 - a) Divisible by at least one of 5, 6, 8?
 - b) Divisible by none of 5, 6, 8?
6. In how many ways 5 number of a's, 4 number of b's and 3 number of c's can be arranged so that all the identical letters are not in a single block?

7. In how many ways can the 26 letters of the English alphabet be permuted so that none of
8. the pattern's CAR, DOG, PUN or BYTE occurs?
9. Find the number of permutations of the letters $a, b, c, \dots, x, y, z$ in which none of the patterns *spin*, *game*, *path* or *net* occurs.

III. Catalan Numbers:

1. Define Catalan number. Obtain the number of paths from $(2,1)$ to $(7,6)$ and not rise above the line $y = x - 1$ using the moves $R: (x, y) \rightarrow (x + 1, y)$ and $U: (x, y) \rightarrow (x, y + 1)$.
2. Using the moves $R: (x, y) \rightarrow (x + 1, y)$ and $U: (x, y) \rightarrow (x, y + 1)$. Find in how many ways can one go from $(2,6)$ to $(6,10)$ and not rise above the line $y = x + 4$.
3. Using the moves $R: (x, y) \rightarrow (x + 1, y)$ and $U: (x, y) \rightarrow (x, y + 1)$. Find in how many ways can one go from $(7,3)$ to $(10,6)$ and not rise above the line $y = x - 4$.
4. Using the moves $R: (x, y) \rightarrow (x + 1, y)$ and $U: (x, y) \rightarrow (x, y + 1)$. Find in how many ways can one go from $(3,8)$ to $(11,16)$ and not rise above the line $y = x + 5$.

IV. Derangements :(Nothing is in its Right Place)

1. Find the number of derangements of 1,2,3,4.
2. Evaluate d_5, d_6, d_7, d_8 .
3. While at race track, a person bets on each of the ten horses in a race to come in accordance to how they are favoured. In how many ways can they reach the finish line so that he losses all his bets?
4. For the positive integers 1, 2, 3,..., n there are 11660 derangements where 1,2,3,4,5 appear in the first five positions. What is the value of 'n'?
5. In how many ways can the integers 1, 2, 3,...,10 be arranged in a line so that no even integer is in its natural place?
6. There are n pairs of children's gloves in a box. Each pair is of a different colour. Suppose the right gloves are distributed at random to n children, and thereafter the left gloves are also distributed to them at random. Find the probability that
 - (i) No child gets a matching pair,
 - (ii) Every child gets a matching pair,
 - (iii) Exactly one child gets a matching pair,
 - (iv) At least 2 children get matching pairs.
7. Thirty students take a quiz. Then for the purpose of grading, the teacher asks the students to exchange papers. Find the probability that
 - (i) No one is grading his own paper.
 - (ii) Every student gets his own paper.
 - (iii) Exactly one student gets his own paper.
8. At a restaurant, 10 men hand over their umbrellas to the receptionist. In how many ways can their umbrellas be returned so that

- (i) No man receives his own umbrella?
- (ii) At least one of the men receives his own umbrella?
- (iii) At least two of the men receives his own umbrella?
